

PROJECT AND TEAM INFORMATION

Project Title

Text Editor

Student / Team Information

<i>Team Name:</i> <i>Team #</i>	<i>DSCPP-III-2026-T185</i>
Team member 1 (Team Lead) <i>(Last Name, name: student ID: email, picture):</i>	<i>Joshi, Arinjay –</i> <i>2510012368</i> <i>2510012268@geu.ac.in</i> 
Team member 2 <i>(Last Name, name: student ID: email, picture):</i>	<i>Mandal, Arunabha –</i> <i>2510011661</i> <i>2510011661@geu.ac.in</i> 
Team member 3 <i>(Last Name, name: student ID: email, picture):</i>	<i>Datta, Mehak –</i> <i>2510011659</i> <i>2510011659@geu.ac.in</i> 
Team member 4 <i>(Last Name, name: student ID: email, picture):</i>	<i>Dastidar, Mehuli –</i> <i>251001160</i> <i>2510011660@geu.ac.in</i> 

PROPOSAL DESCRIPTION (10 pts)

Motivation (1 pt)

We picked a project for our learning team: "Text Editor Using Data Structures and C++." Our goal was not just to build something. We wanted to understand how a text editor works under the hood. We use one every day, yet we rarely think about what makes typing, changes, and fixes happen. We plan to study the basic actions. This includes inserting text, deleting text, searching for words, replacing text, and also undo and redo. We will implement these tasks with data structures like linked lists and stacks.

Another reason we chose this topic is to connect what we learned in class to real life. While we work, we will use ideas such as classes and objects, along with encapsulation and abstraction. We will also practice constructors, pointers, functions, and file handling. This helps the theory feel less distant. We also expect the coding itself to teach us. Since we will write the application, we will face errors and bugs. Then we will need to debug and adjust the program until it works.

In the end, our main motivation is to get a clearer feel for data structures and object oriented programming by making a tool that can be used in everyday life.

State of the Art / Current solution (1 pt)

We chose this project because we use a text editor every day, yet we rarely pause to think about how it actually works. I mean the basic steps behind it. We want to learn how features like insert, delete, search, replace, undo, and redo are done under the hood. We plan to study how data structures, such as linked lists and stacks, support those features.

Another reason for our choice was simple. We wanted the ideas from class to match real situations. So we set out to connect what we learn with something we see and use often.

While working on the project, we expected to grow our understanding of core terms. These include classes, objects, encapsulation, abstraction, constructors, pointers, functions, and file handling. We also thought it would help our debugging habits and our coding ability. Since we would build the application ourselves, we knew we would run into a fair number of issues, and we would have to deal with them.

Overall, our main goal is to get more comfortable with data structures and object oriented programming. We do this by building an application that can help in everyday life.

Project Goals and Milestones (2 pts)

The essential objective of this project is to develop a console-based text editing software with the aid of Data Structures along with C++. This text editor will be developed in a way that it will have all the important features that are required in a simple text editor such as insertion of texts, deletion of texts, movement of the cursor, features such as undo and redo, copy and paste feature, saving and loading of texts. In the course of developing this project, we will learn about the applications of linked list, stacks, files and OOP. Our project is going to be implemented through the following approaches. First, we have to learn and understand the way in which the text editor appears. Secondly, we have to define the way in which the text, cursor, and the clipboard will appear in our text editor. As such, we will go ahead to the designing and implementation phase. This will entail creating texts, moving of the cursor and insertion and deletion of texts. The other phases will include implementation of the undo and redo commands using stack, as well as the search function. The last step will be testing of the software.

Project Approach (3 pts)

We would like to develop this editor by analyzing the basic operations which should be expected from the software and this includes creating the text, inserting or deleting any information, modifying or editing text, searching for any word and saving the file. Rather than implementing these operations through some pre-defined text editor or built-in data structures only, we will be implementing the basic operations on our own using data structures.

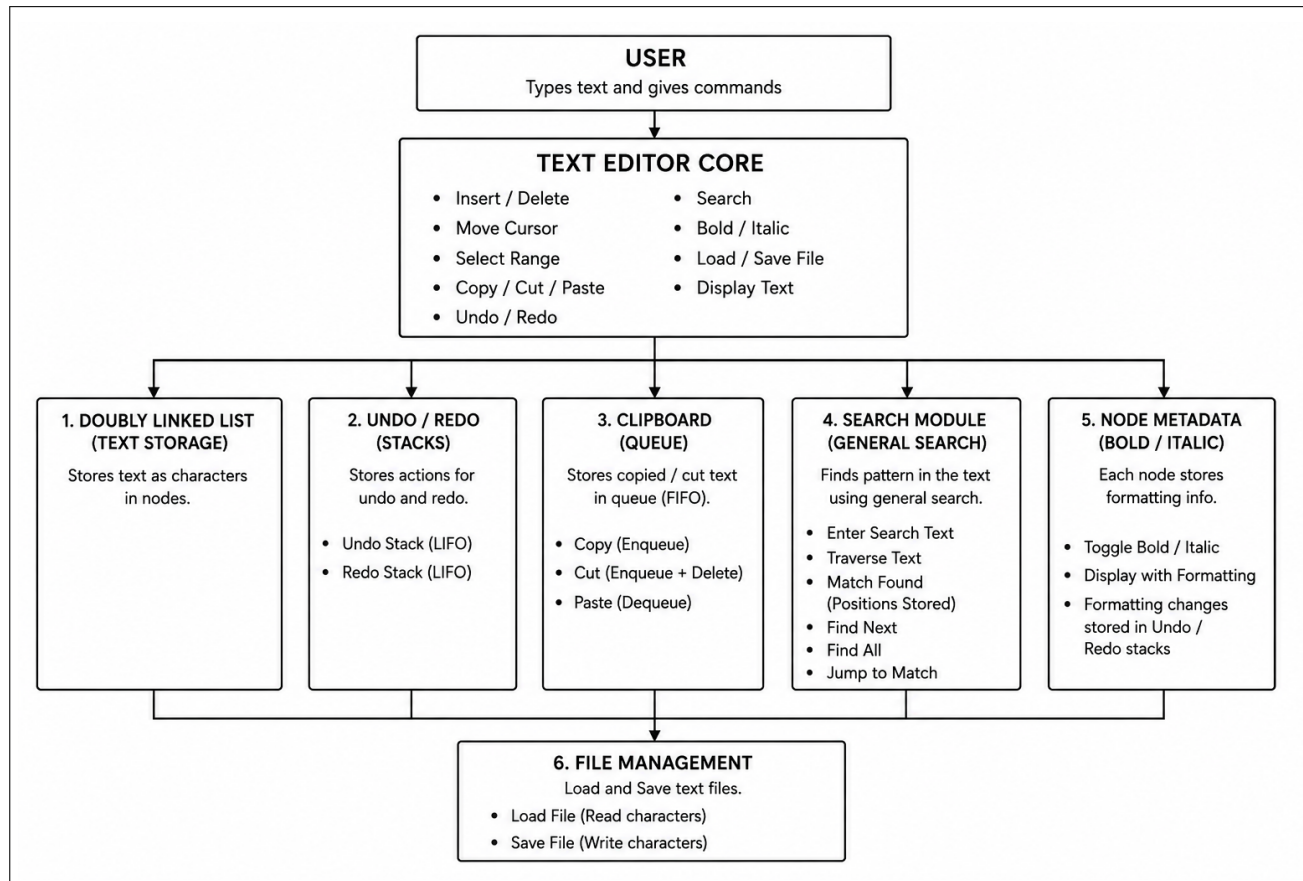
C++ programming language is going to be used in this entire project as it provides us the liberty to work with various types of data structures along with object-oriented programming. The appropriate data structure will be selected wherever necessary such as linked list, stack, and other basic data structures.

For development, we will use a Windows-based computer with an IDE/code editor such as Visual Studio Code or Code::Blocks, along with a C++ compiler. The project will primarily be console-based, which will keep the interface simple and allow us to focus more on the actual implementation of the data structures and algorithms.

We will build the project step by step, starting with the basic text-handling functions and then adding more features. Finally, we will test each operation individually and then test the complete system to make sure all the features work together properly.

Our overall approach is to keep the design simple, modular, and easy to understand, while demonstrating how Data Structures can be used to create a practical application.

System Architecture (High Level Diagram)(2 pts)



Project Outcome / Deliverables (1 pts)

The main outcome of our project will be a working text editor developed using C++ and Data Structures. The editor will provide basic features that a user would expect, such as creating and managing text, inserting and deleting content, editing text, searching for specific words, and saving files.

Another important outcome is the practical implementation of the Data Structures and Object-Oriented Programming concepts that we have studied. Apart from learning about these concepts in theory, we will apply them to develop real-life features of the text editor, which will help us get a better idea of data storage, manipulation, retrieval, and searching in applications.

Major deliverables of this project will be:

C++ Text Editor with working features. Data structures implementation to store and process text. Features for inserting/deleting/editing/searching text. File handling feature to open/save text files. Well-structured and properly documented source code. Testing and debugging of developed features. Project Report describing the strategy used, data structures used, implementation, and outcomes of the project. Demonstration of the project.

In conclusion, this project will provide a working application along with the demonstration of our knowledge on Data Structures, Algorithms, and C++.

Assumptions

In designing the text editor, it is assumed that it would only be used in editing simple texts and not complex ones. It is assumed that the user will be having basic knowledge of using a text editor. The program will be executed on a computer which runs Windows Operating system and C++ Compiler. The inputs will include text data and no large size files. The data structure that will be used is supposed to be sufficient for executing the required operations.

References

1. Bjarne Stroustrup, *The C++ Programming Language, 4th Edition*, Addison-Wesley.
2. Thomas H. Cormen et al., *Introduction to Algorithms, 4th Edition*, MIT Press.
3. GeeksforGeeks, *C++ Data Structures and Algorithms*.
4. *C++ Reference, Standard Library Documentation*.
5. *ISO C++, C++ Standard Library Documentation*.